

Al-Naser Al-Rais

Researching brain ageing and neurodegeneration
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EDUCATION

University of Cambridge

BA in Natural Sciences (Hons)

- **Courses:** Biochemistry, Computational Biology, Pharmacology, Mathematical Biology, Chemistry
- **Specialization:** Genetics

Cambridge, UK

Oct 2024 - June 2027

EXPERIENCE

Church Lab, Wyss Institute, Harvard (Incoming)

Undergraduate Research Fellow,

- Partial reprogramming to reverse brain aging in dopaminergic neurons for Parkinson's Disease
- Self-funded via independent fundraising and Vivensa Foundation Ignition Award (\$15k)

Boston, MA

Jun 2026 – Sep 2026

Knowles Lab, University of Cambridge

Undergraduate Researcher, Chemistry

- Fabricated PDMS microfluidic devices using soft lithography for peptide hydrogel screening
- Built UV spectroscopy calibration curves and measured peptide hydrogel dissociation kinetics across pH/salt conditions
- Investigated full length-Tau monomer states (coil-like vs premolten globule-like) using CD spectroscopy
- Explored microelectrophoresis approaches for single-particle charge determination of LNPs and condensates

Cambridge, UK

Oct 2025 – Present

Lu Lab, Harvard Medical School

Undergraduate Research Fellow, Systems Biology

- Engineered doxycycline-inducible protein aggregation model in HEK293T and U2OS cells
- Expanded model to include Tau, alpha-synuclein, TDP-43, FUS via Gibson assembly
- Generated stable cell lines expressing sfGFP-tagged constructs using lentiviral transduction
- Created 20 P301X Tau (K18 domain) variants using site-directed mutagenesis to map aggregation propensity and toxicity
- Quantified aggregate formation and cytotoxicity using fluorescence microscopy and cell detachment counts

Boston, MA

Jun 2025 – Sep 2025

HONOURS & AWARDS

Secured \$8000 from angel investors to fund placement at Church Lab, Harvard	2026
Vivensa Foundation Ignition Award - \$7000 for aging research in Church Lab, Harvard	2026
G.R.N. Minchin Award, University of Cambridge - Substantial travel grant	2025
TU Munich / Max Planck Placement - 1 of 10 selected worldwide, fully funded	2025
Bio-Spark Exploring Fellow	2025

EXTRACURRICULAR ACTIVITIES

Cambridge Society for Ageing Research – Head of Undergraduate Engagement

- Organised 100-person screening of *Forever Young*, hosting Prof. Steve Horvath (Altos Labs)

Jan 2026 – Present

No Patient Left Behind – Fellow

- Authored commentary on NICE's rejection of lecanemab/donanemab (on website)

Mar 2026 – May 2026

Darwin Society (Christ's College Scientific Society) – Speaker's Officer

- Interviewed Sir Mene Pangalos (Venture Partner, Omega Funds); 5 speaker events; 3 formals

Jun 2025 – Jun 2026

British Powerlifting – National Competitor

2022